TechNova 2026
Mathsmania Advance
Sample Paper – Round 1

Department of Mathematics and Statistics
Institute of Business Management (IoBM), Karachi

Duration:	60 Minutes	**Total Questions:**	40
Marks:	40	**Negative Marking:**	None

INSTRUCTIONS TO CANDIDATES

- Attempt all questions. Each question carries exactly one mark.
- There is no negative marking for incorrect answers.
- All questions are Multiple Choice Questions (MCQs) with four options and a single correct response.
- Calculators and any other electronic devices are strictly prohibited.
- Use clean, legible rough work spaces where necessary. Do not write anything on the question booklet except your details.
- Ensure that you mark your final answers clearly on the provided response sheet.

SECTION A: ALGEBRA (Questions 1–12)

Question 1: If the roots of the quadratic equation $x^2 - bx + c = 0$ are two consecutive integers, then $b^2 - 4c$ is equal to:

A) 0 B) 1 C) 2 D) 4

Question 2: Let $f(x) = (x - 1)(x - 2)(x - 3)...(x - 2026)$. The coefficient of x^{2025} in the expanded polynomial is:

A) -2053325 B) -2051351 C) 2051351 D) 0

Question 3: Find the sum of all real solutions to the equation $|x^2 - 3x - 4| = 2$.

A) 3 B) 6 C) 0 D) 9

Question 4: If the system of linear equations $x + y + z = 2$, $2x + 3y + 2z = 5$, and $2x + 3y + (a^2 - 1)z = a + 3$ has infinitely many solutions, then the value of a is:

A) 2 B) -2 C) $\sqrt{5}$ D) $-\sqrt{5}$

Question 5: The solution set of the inequality $(x - 1)^2(x + 2) / (x - 4) \leq 0$ is:

A) [-2, 4) B) [-2, 1] ∪ (1, 4) C) (-∞, -2] D) [-2, 4]

Question 6: If the sum of the first n terms of an arithmetic progression is given by $S_n = 3n^2 + 5n$, then the 10th term of this progression is:

A) 56 B) 62 C) 68 D) 74

Question 7: Let $f: R \to R$ be a function satisfying $f(x + y) = f(x)f(y)$ for all $x, y \in R$. If $f(1) = 3$, then the value of Σ (from k=1 to 4) f(k) is:

A) 39 B) 81 C) 120 D) 360

Question 8: The domain of the real-valued function $f(x) = \sqrt{\log_{0.5}(x - 2)}$ is:

A) (2, 3] B) (2, ∞) C) [3, ∞) D) (2, 2.5]

Question 9: If the function $f(x) = (ax + b) / (cx + d)$ is its own inverse, where a, b, c, d are non-zero real numbers, then which of the following must be true?

A) $a = d$ B) $a = -d$ C) $b = c$ D) $b = -c$

Question 10: The geometric mean of two positive numbers a and b is 8, and their arithmetic mean is 10. The absolute difference between the numbers is:

A) 6 B) 8 C) 12 D) 16

Question 11: The fractional part of any real number x is denoted by $\{x\} = x - \lfloor x \rfloor$. Solve the equation $3\lfloor x \rfloor + 2\{x\} = 5.4$. The value of x is:

A) 1.2 B) 1.7 C) 2.2 D) 2.7

Question 12: If $x + 1/x = 3$, then the value of $x^4 + 1/x^4$ is:

A) 47 B) 49 C) 7 D) 81

SECTION B: NUMBER THEORY (Questions 13–20)

Question 13: What is the remainder when 3^{2026} is divided by 7?

A) 1 B) 2 C) 3 D) 4

Question 14: The number of positive integral divisors of 2026 is (Note: $2026 = 2 \times 1013$, where 1013 is prime):

A) 2 B) 4 C) 6 D) 8

Question 15: If the greatest common divisor of two positive integers a and b is 12, and their least common multiple is 720, how many pairs (a, b) are possible?

A) 4 B) 8 C) 16 D) 32

Question 16: Let n be a positive integer such that n! ends in exactly 100 zeros when written in base 10. The smallest possible value of n is:

A) 400 B) 405 C) 410 D) 415

Question 17: If a and b are integers such that 7a + 3b is divisible by 11, then which of the following is always divisible by 11?

A) a + 2b B) 3a - 4b C) 4a - 2b D) 2a + 4b

Question 18: The last digit of the sum 1! + 2! + 3! + ... + 2026! is:

A) 1 B) 3 C) 5 D) 7

Question 19: The number of ordered pairs of positive integers (x, y) that satisfy the equation 1/x + 1/y = 1/6 is:

A) 5 B) 9 C) 12 D) 15

Question 20: The product of four consecutive integers is always divisible by which of the following numbers?

A) 12 B) 24 C) 48 D) 96

SECTION C: STATISTICS & PROBABILITY (Questions 21–30)

Question 21: The mean of 10 observations is 25. If one observation of value 34 is removed, the mean of the remaining observations becomes:

A) 23 B) 24 C) 25 D) 26

Question 22: A dataset consists of 11 distinct positive integers. If the largest integer is increased by 20, which of the following descriptive statistics will change?

A) Median B) Mode C) Mean D) Interquartile Range

Question 23: The standard deviation of a set of 50 observations is 8. If each observation is multiplied by -3, the new standard deviation is:

A) 24 B) -24 C) 8 D) 64

Question 24: Two fair six-sided dice are rolled simultaneously. What is the probability that the sum of the top faces is a prime number?

A) 5/12 B) 7/18 C) 1/3 D) 11/36

Question 25: A box contains 5 red, 4 blue, and 3 green balls. If 3 balls are drawn at random without replacement, the probability that all three are of different colors is:

A) 3/11 B) 4/11 C) 5/22 D) 6/11

Question 26: How many distinct 5-letter arrangements can be made using the letters of the word 'TECHNOVA' such that the vowels always occupy the first and last positions?

A) 360 B) 720 C) 120 D) 1440

Question 27: If $P(A) = 0.6$, $P(B) = 0.4$, and $P(A \cup B) = 0.8$, then the conditional probability $P(A \mid B)$ is equal to:

A) 0.2 B) 0.4 C) 0.5 D) 0.6

Question 28: In a certain university competition, the probability that team A wins is 1/3 and the probability that team B wins is 1/4. Assuming a draw is impossible, the probability that either team A or team B wins is:

A) 7/12 B) 1/12 C) 1/2 D) 5/12

Question 29: The variance of the first n natural numbers is given by which of the following expressions?

A) $(n^2 - 1) / 12$ B) $(n^2 - 1) / 6$ C) $(n + 1)(2n + 1) / 6$ D) $n^2 / 4$

Question 30: A student is selecting 3 courses out of 7 available electives. If 2 specific courses cannot be taken together, how many valid combinations of 3 courses can the student choose?

A) 21 B) 30 C) 35 D) 25

SECTION D: CALCULUS (Questions 31–40)

Question 31: Evaluate the limit: $\lim (x \to 0) (\tan(x) - \sin(x)) / x^3$.

A) 0 B) 1/2 C) 1 D) Does not exist

Question 32: If $f(x) = x^{\wedge}(x)$, then the value of the derivative $f'(1)$ is:

A) 0 B) 1 C) e D) ln(2)

Question 33: The maximum value of the function $f(x) = x^3 - 3x$ on the closed interval $[-1, 2]$ is:

A) 2 B) 0 C) -2 D) 4

Question 34: The absolute minimum value of the function $g(x) = x + 4/x$ for $x > 0$ is:

A) 2 B) 4 C) 8 D) 0

Question 35: Evaluate the definite integral: $\int$ (from 0 to $\pi/2$) $\sin^3(x) \cos(x)\, dx$.

A) 1/4 B) 1/3 C) 1/2 D) 1

Question 36: If $F(x) = \int$ (from 1 to x^2) $\sqrt{t^2 + 3}\, dt$, then the value of the derivative $F'(2)$ is:

A) $2\sqrt{7}$ B) $4\sqrt{19}$ C) $2\sqrt{19}$ D) $4\sqrt{7}$

Question 37: The total area bounded by the curve $y = x^2 - 4$ and the x-axis from $x = -2$ to $x = 2$ is:

A) 16/3 B) 32/3 C) 8 D) 0

Question 38: A rectangular field is to be fenced against an existing straight wall. If 100 meters of fencing material is available, what is the maximum area that can be enclosed?

A) 625 m^2 B) 1250 m^2 C) 2500 m^2 D) 500 m^2

Question 39: The function $f(x) = 2x^3 - 9x^2 + 12x - 3$ is strictly decreasing in the interval:

A) (1, 2) B) $(-\infty, 1)$ C) $(2, \infty)$ D) (1, 3)

Question 40: Evaluate the limit: lim $(x \to \infty)$ $(1 + 2/x)$^x.

A) e B) e^2 C) 1 D) ∞